Prepared for future scientists

Gauss-Laplace Synthesis

A brief review of the birth of normal distribution and central limit theorem

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Abstract: This manuscript provides a brief retrospective on the emergence of the normal distribution and the central limit theorem in history, highlighting the contributions of C. F. Gauss and P. -S. Laplace. For the convenience of the readers, we do not follow the exact treatments in their original works, but provide simplified versions that preserve the spirit. During the preparation of this manuscript, we also notice a few common misconceptions in some textbooks of statistics that may lead to confusion, which are pointed out and accompanied by correct reasoning or derivations. We hope this short note can help readers gain a better understanding in related subjects like error analysis and statistical inference.

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1 Prologue

When introducing the normal distribution, most textbooks start with the pioneering work [1] by Abraham de Moivre (1667-1754) in 1738, which introduced the concept of normal distribution as approximations to binomial distributions when the number of trials $n \to \infty$.

The statement of de Moivre's result is as follows:

Theorem 1.1 (de Moivre-Laplace theorem)**.** *The probability mass function (分布列) of the random number of "successes" observed in a series of n independent Bernoulli trials, each having probability p of success (or in short, a binomial distribution of n trials $B(n,p)$), converges to the probability density function of the normal distribution with expectation np and standard deviation $\sqrt{np(1-p)}$, as n grows large, assuming $p \neq 0, 1$.*

More precisely speaking, for a random variable $X \sim B(n,p)$, its probability mass function

$$p_X(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \tag{1.1}$$

becomes approximately

$$P(X = k) \simeq \frac{1}{\sqrt{2\pi np(1-p)}} e^{-\frac{(k-np)^2}{2np(1-p)}} \tag{1.2}$$

as n grows large, in the sense that the ratio of the left-hand side (LHS) and the right-hand side (RHS) converges to 1 as $n \to \infty$[1]. This result is so classic that it is often demonstrated in courses of probability and statistics using Galton's bean machine or numerical computation.

De Moivre's result is usually seen as a special case of the central limit theorem[2] published by Pierre-Simon Laplace (1749-1827) in 1812 in his influential book *Théorie analytique des probabilités* [2]. This is why de Moivre's result, though independently published by himself more than 70 years earlier than Laplace, is often referred to as "de Moivre-Laplace theorem". Following this spirit, many textbooks use de Moivre's result to *demonstrate* central limit theorem, without even bothering to provide a proof. However, due to heavy computations involved in deriving de Moivre's result and its weak connection with the central limit theorem, students usually don't get the essence of all these topics, not to mention appreciating their beauty and power.

[1] Strictly speaking, this approximation (known as the Local Limit Theorem) is uniformly valid only when k is in the $\mathcal{O}(\sqrt{n})$ neighborhood of the mean np. In the extreme tails where $|k - np|$ scales with n, the ratio between the exact probability and the Gaussian approximation diverges, entering the realm of Large Deviation Theory.

[2] If you have never heard of what it is, take it for now as a "very important theorem" of probability theory and statistics, and continue reading –discussing the central limit theorem is one of the main purposes of this manuscript.

This manuscript provides a more transparent, history-based introduction to normal distribution and the central limit theorem by extracting the main path taken successively by Carl Friedrich Gauss (1777-1855) and Laplace. This epic was named **Gauss-Laplace synthesis** by the eminent statistician and historian of science Stephen M. Stigler in his landmark 1990 book [3], where it serves as the title of Chapter 4, marking the transition of the subject of statistics from empiricism to science supported by rigorous mathematical foundations.

Disclaimer: For simplicity, we will not follow the exact treatments in Gauss and Laplace's original works, but instead largely reducing the complexity by considering less complicated problems and with the help of modern mathematics. Therefore, readers should not take this manuscript as an authentic review of the original recipe, but a rendition with the same spirit. For example, when we say something like "Gauss eventually found a way to tackle this problem", what actually happened in history can be that Gauss used a not very similar method to solve a problem with a totally different form. With this being said at the beginning, we will not mention too much about the similarity and difference of our treatments and the actual history along the way, in order not to distract the readers from their mind flow.

2 Arithmetic mean and maximum likelihood (Gauss 1809)

2.1 Arithmetic mean and least squares

We all learned from elementary school that all observations and measurements in the real world contain inevitable random errors. To fight against such uncertainty and get better estimate of the "true value", a widely used strategy is that we measure the same quantity several times, and use the arithmetic mean of all the observations as a better estimate. A rough intuition for using arithmetic mean is that the random errors contained in different observations tend to cancel out when we sum them together, since the positive and negative errors are "equally probable" due to its random nature.

However, when one tries to think harder on this idea, severe problems start arising, questioning whether taking arithmetic mean is the only reasonable strategy, or at least more reasonable than others[3]. What is so special about the arithmetic mean? In 1801, when Gauss was working on extrapolating the motion of Ceres[4] based on very limited amount of observational data, he realized that to justify the arithmetic mean, or anything else, to be the "best" estimate, there needs to be a measure of "how good an estimate is based on the observed data". Such a measure is called a **loss function** in modern terminology. For n times of independent measurements of a quantity that return n observations $X_1 = x_1, \ldots, X_n = x_n$, the loss function proposed by Gauss was

$$L(\hat{\mu}) = \sum_{i=1}^{n} (x_i - \hat{\mu})^2, \tag{2.1}$$

where $\hat{\mu}$ is an estimate of the true value of the quantity, which is a function of the values of the observations $x_1, \ldots, x_n$. Frankly speaking, $L(\hat{\mu})$ measures how far away the estimate $\hat{\mu}$ is from all the observations. The larger $L(\hat{\mu})$ is, the further away $\hat{\mu}$ is from the data and it is regarded as a worse estimate.

So clearly, according to Gauss's loss function eq. (2.1), the "best" estimate $\hat{\mu}$ should minimize $L(\hat{\mu})$. Since the loss function $L(\hat{\mu})$ is the sum of the squares of the deviation between values of observations x_i's and $\hat{\mu}$, the method of determining best $\hat{\mu}$ by minimizing $L(\hat{\mu})$ bears the name **least squares**[5]. By requiring the derivative

$$L'(\mu) = -2 \sum_{i=1}^{n} (x_i - \hat{\mu}) = 0, \tag{2.2}$$

we get the best estimate to be

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} x_i, \tag{2.3}$$

[3] For instance, by taking mean of the cube of every observation, and then taking a cubic root, is also reasonable by the simple arguments that positive and negative errors tend to cancel out.

[4] A dwarf planet in the main asteroid belt between the orbits of Mars and Jupiter.

[5] Based on the same philosophy, Gauss developed the least squares method for curve fitting, which he immediately applied to the Ceres problem. We will not elaborate it in this manuscript.

which is exactly the arithmetic mean of all the observations.

Wait, it does not explain anything! What we showed just now is that the arithmetic mean is the best estimate in the sense that it minimizes the loss function eq. (2.1) which Gauss proposed, but why the loss function has to take that specific form in the first place? Gauss needed to find an argument for that.

2.2 Arithmetic mean and maximum likelihood

Gauss proceeded in the following way: Every observation X_i can be thought as

$$X_i = \mu + \delta_i, \quad i = 1, \ldots, n \tag{2.4}$$

where μ is the objective "true value" of the quantity unknown to us and δ_i denotes the random error obtained in the i^{th} observation. Since δ_i is a continuous random variable, it has a probability density function (PDF) $f(x)$, which is the same for $\delta_1, \ldots, \delta_n$. Equivalently, it means that every observation X_i, when seen as a random variable, has a PDF

$$f_{X_i}(x) = f(x - \mu). \tag{2.5}$$

Therefore, the joint PDF of all the n observations $\boldsymbol{X} = (X_1, \ldots, X_n)$ is the product

$$f_{\boldsymbol{X}}(x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i - \mu). \tag{2.6}$$

Here comes Gauss's marvelous idea. He noticed that the usual way of understanding eq. (2.6) is to think the true value μ to be a fixed constant, while regarding $x_1, \ldots, x_n$ as variables, reflecting the objectiveness of μ and the randomness of every X_i. However, from the perspective of the observer, it is the recorded data $x_1, \ldots, x_n$ that is actually fixed, while the seemingly objective μ is unknown. To highlight this idea, Gauss replaced the true value μ with its estimate $\hat{\mu}$ in the joint PDF eq. (2.6), reinterpreting it as a function of $\hat{\mu}$ called **likelihood function**

$$\mathcal{L}(\hat{\mu}) = \prod_{i=1}^{n} f(x_i - \hat{\mu}), \tag{2.7}$$

bearing the meaning "the probability density for the n independent observations to return the specific $(x_1, \ldots, x_n)$ *when the true value equals* $\hat{\mu}$". This is to say, treating the estimate $\hat{\mu}$ as a parameter, the observation result $(x_1, \ldots, x_n)$ is more likely to happen for some values of $\hat{\mu}$, while being less likely to happen for some other values of $\hat{\mu}$ –this likelihood is measured by the likelihood function $\mathcal{L}(\hat{\mu})$.

Gauss's intention might be clear to the readers now: the best estimate $\hat{\mu}$ should maximize the likelihood $\mathcal{L}(\hat{\mu})$. His reasoning[6] is that the actually observed data $(x_1, \ldots, x_n)$

[6] It is actually more like an intuition, or an idea, instead of a rigorous reasoning.

should be very likely to pop up provided the true μ, so the $\hat{\mu}$ that maximizes the likelihood should be a good estimate of μ[7].

Though the idea is great, there is no way to go ahead evaluating the best $\hat{\mu}$ since we don't know the form of the error PDF $f(x)$ in eq. (2.7). However, we shall recall that our aim is to justify the use of the arithmetic mean as the best estimate: We have made a good argument that maximum likelihood is a good way of defining the best estimate, so a possible way to proceed is to *find* the specific $f(x)$ such that the arithmetic mean $\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$ actually maximizes eq. (2.7).

A generic math problem of this kind is rather hard to solve, because formally speaking, $\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$ is a fixed number now and we are treating $\mathcal{L}(\hat{\mu})$ as a functional of the function $f(x)$, thus maximizing it by finding a suitable $f(x)$ is a problem of variational calculus.

However, Gauss, the *princeps mathematicorum*, did not try to tackle this problem in the usual brute-force way. He firstly took a logarithm of eq. (2.7)[8], resulting in what we call **log-likelihood** today:

$$\ell(\hat{\mu}) = \log[\mathcal{L}(\hat{\mu})] = \sum_{i=1}^{n} \log[f(x_i - \hat{\mu})]. \tag{2.8}$$

Since logarithm is strictly increasing, the log-likelihood $\ell(\hat{\mu})$ has the same extrema as $\mathcal{L}(\hat{\mu})$. So the $\hat{\mu}$ maximizing $\mathcal{L}(\hat{\mu})$ satisfies

$$\ell'(\hat{\mu}) = -\sum_{i=1}^{n} \frac{f'(x_i - \hat{\mu})}{f(x_i - \hat{\mu})} = 0. \tag{2.9}$$

Denoting $g(x) = f'(x)/f(x)$, it becomes

$$\ell'(\hat{\mu}) = -\sum_{i=1}^{n} g(x_i - \hat{\mu}) = 0, \tag{2.10}$$

a formal equation for best estimate $\hat{\mu}$ which maximizes the likelihood function $\mathcal{L}(\hat{\mu})$.

The above treatment may seem useless at first glance since it does nothing but transforming the problem of maximizing a functional $\mathcal{L}(\hat{\mu})$ of $f(x)$ into the problem of finding the zero of another functional $\ell'(\hat{\mu})$ of $g(x)$, but Gauss acutely noticed that the equation eq. (2.10) for determining the maximum likelihood $\hat{\mu}$ looks very similar to the equation

[7]In modern statistics, what we just did can be seen as estimating an unknown parameter μ of an assumed probability distribution eq. (2.5), given some observed data $(x_1, \ldots, x_n)$. This is called **statistical inference**, i.e., inferring useful information from statistical data. Gauss's method of estimating parameters by maximizing a likelihood function, called **maximum likelihood estimation**, eventually becomes very famous and widely applied.

[8]When investigating a function of a product form like eq. (2.7), a standard treatment is to take a logarithm to transform the product into a summation, for further convenience.

eq. (2.2) for determining the least squares $\hat{\mu}$. Precisely speaking, eq. (2.2) is a special case of eq. (2.10) where $g(x) = 2x$. Since we already know that the arithmetic mean $\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$ of the n observations is the solution of eq. (2.2), $g(x) = ax$ for any (real) a will make $\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$ to satisfy eq. (2.10). A zero of the functional $\ell'(\hat{\mu})$ of $g(x)$ is found!

According to $g(x) = f'(x)/f(x) = (\log[f(x)])'$, the above result $g(x) = ax$ indicates

$$\log[f(x)] = \int g(x)dx = \frac{1}{2}ax^2 + b, \tag{2.11}$$

which means

$$f(x) = e^{\frac{1}{2}ax^2+b} = Ce^{\frac{1}{2}ax^2}. \tag{2.12}$$

Recall that $f(x)$ is the PDF of every error δ_i, it has to be normalized in the sense $\int f(x)dx = 1$, from which we know that $a < 0$ (otherwise not normalizable) and C is a proper normalization constant determined by a. Rewriting $a = -\frac{1}{2\sigma^2}$, we get

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{x^2}{2\sigma^2}}. \tag{2.13}$$

This PDF of random error is the profound discovery by Gauss in 1809 [4]. Let's give it a formal statement:

Proposition 2.1. *For n independent observations $X_1 = x_1, \ldots, X_n = x_n$ of a quantity, the arithmetic mean $\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$ is the maximum likelihood estimation of the true value μ* ***if*** *the random errors δ_i's have the PDF of form eq. (2.13).*

In eq. (2.13), the parameter σ can take any real value without affecting the conclusion in theorem 2.1, but it is conventionally taken to be positive. The reason is that by doing a not-so-easy integral, one can show that the variance of an error δ_i

$$\mathbb{V}(\delta_i) = \int_{-\infty}^{+\infty} x^2 f(x)dx = \sigma^2, \tag{2.14}$$

and people usually like to regard σ as the standard deviation. The bell-shaped function eq. (2.13), or more precisely its generic form (actually nothing but shifted to $+x$ by μ)

$$\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \tag{2.15}$$

is gradually found to play a variety of key roles in the theory of probability, statistics, and other branches of mathematics and sciences, which get it a bunch of names. When playing the role of a PDF, it is named **central, Gaussian, or normal distribution**, and a random variable X having such a PDF will be denoted by $X \sim \mathcal{N}(\mu, \sigma^2)$; when being a function, it is called Gaussian (or Gauss) function, or simply "bell curve". The statistical and analytical properties of it can be found in almost every textbook of statistics (even some of physics and chemistry).

Well, before we celebrate Gauss's success in showing theorem 2.1, it is worth noticing that theorem 2.1 only includes the "if" part of the statement[9]. Actually, its converse, the "only if" part, is also true. To prove it, one only needs to find some special cases of the observations $(x_1, \ldots, x_n)$ to exclude other possible forms of $f(x)$. We leave it as an exercise to the readers[10]. Therefore, theorem 2.1 and its converse unifies to a theorem:

Theorem 2.2. *In a series of independent observations* $X_1 = x_1, \ldots, X_n = x_n$*, their arithmetic mean* $\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$ *is the maximum likelihood estimation of the true value* μ ***if and only if*** *the random error* δ_i *in every observation* $X_i = \mu + \delta_i$ *has zero expectation* $\mathbb{E}(\delta_i) = 0$ *and is normally distributed, that is,* $\delta_i \sim \mathcal{N}(0, \sigma^2)$.

This concludes Gauss's contribution to the rise of normal distribution and, indirectly, the central limit theorem. Loosely speaking, he showed that if we want to justify the arithmetic mean of a series independent observations as the "best" (in the maximum likelihood sense) estimation of the true value, we have to count on a fact that the random errors are normally distributed. But in our physical world, is it really the case that random errors in observations are always, or at least mostly, normally distributed? If so, what is the mechanism for such a universal consequence? If not, shall we update our strategy of estimation from using the arithmetic mean to something else for improvement? These questions were not answered by Gauss in his 1809 paper, however, Gauss's work and these unresolved questions greatly inspired Laplace, a sixty-year-old renowned mathematician and astronomer who has been working on this field of study in most of his life, whose follow-up work published 3 years later promoted normal distribution to its zenith in statistics.

3 From epistemology to ontology: the central limit theorem (Laplace 1810-1812)

From the philosophical point of view, Gauss's discovery of normal distribution (mainly theorem 2.2) is of an **epistemological** (认识论的) fashion: it does not concern about *why* the mother nature generates normally distributed random errors. Instead, provided the *experience* that the arithmetic mean usually does well in estimations, Gauss provided an understanding –the observations are *most likely* to appear when the true value equals to their arithmetic mean –and derived theorem 2.2, the sufficient and necessary condition for his understanding to be correct. Studies of such epistemological fashion are widely seen in different fields of science, for instance, using maximum entropy to understand thermal equilibrium, though they often exude some tinge of half-measures.

[9] That is, eq. (2.13) is *a* PDF of the random errors that can make the arithmetic mean to be the maximum likelihood estimation, but may not be *the only* PDF that can do the job.

[10] Solution: For instance, let's take $x_1 = n\hat{\mu}$ and $x_2 = \cdots = x_n = 0$, where $\hat{\mu}$ really is the arithmetic mean of the n observations. Then eq. (2.10) becomes $g((n-1)\hat{\mu}) + (n-1)g(-\hat{\mu}) = 0$, which should be valid for $n = 1, 2, 3, \ldots$. The $n = 2$ case yields $g(\hat{\mu}) + g(-\hat{\mu}) = 0$, so that $g(x)$ is an odd function. This reduces $g((n-1)\hat{\mu}) + (n-1)g(-\hat{\mu}) = 0$ to $g((n-1)\hat{\mu}) = (n-1)g(\hat{\mu})$, from which one can see $g(x)$ is linear in x: $g(x) = ax$ for arbitrary a.

In contrast to epistemological approaches, **ontological** (本体论的) approaches in scientific studies, which aim for providing a theory or model[11] to *explain* the underlying mechanism of what we see, are often more satisfactory, at least to the general audience. In our present story about normal distribution, it was Laplace who took over Gauss's position, providing an ontological understanding of the distribution of random errors [2, 5][12].

3.1 Composition of many error sources

In the entire section 3 we forget about the n independent observations and the errors in them, but to focus on the error δ of one specific observation. Laplace noticed that in reality, the random error δ almost always comes from a variety of sources, such as small vibrations of the apparatus, random fluctuations of the density/temperature of gas/liquid, and random noises that can never be fully screened. Laplace perceived that δ should consist of the contributions from all the sources:

$$\delta = \epsilon_1 + \ldots + \epsilon_N = \sum_{i=1}^{N} \epsilon_i, \tag{3.1}$$

where ϵ_i denotes the contribution from the i^{th} source, following an unknown distribution $f_i(x)$. Please notice that when saying every ϵ_i has a distribution $f_i(x)$, we silently assume that they are all independent. Laplace further assumed that every ϵ_i has zero expectation

$$\mathbb{E}(\epsilon_i) = 0, \tag{3.2}$$

which is reasonable because whenever $\mathbb{E}(\epsilon_i) = \mu_i \neq 0$, we can redefine $\widetilde{\epsilon}_i = \epsilon_i - \mu_i$, whose expectation is zero[13]. Additionally, let's denote the variance of ϵ_i as

$$\mathbb{V}(\epsilon_i) = \sigma_i^2, \tag{3.3}$$

where $\sigma_i > 0$ is the standard deviation. We do not assume anything about σ_i.

Now that the distribution $f_i(x)$ of every ϵ_i, though hidden from the observer, is completely determined by some complicated underlying physical processes, Laplace's next step is to extract the distribution $f(x)$ of δ based on all the $f_i(x)$'s, trying to make some generic statements about it. In this way, the problem shrinks down to: what is the distribution of δ, which is a sum of a bunch of independent variables $\epsilon_1, \ldots, \epsilon_n$?

[11] In fact, in most cases, these models are so simplified that they remain far from actual reality. However, as long as a model captures the key factors driving the observed phenomena, it offers a sense of conceptual reassurance.

[12] Laplace first submitted his derivations to The Paris Academy of Sciences in 1810 [5], and then provided a systematic treatment of the theory of probability in 1812 [2].

[13] The nonzero component μ_i is regarded as a **systematic error**, in contrast to random errors. Statistical methods can only address random errors, namely the $\widetilde{\epsilon}_i$ term. Systematic errors, on the other hand, need to be eliminated by identifying and subtracting them through a careful investigation of their sources in actual measurements.

To get a sense of the scope of this problem, let's think about a simplest case in which $Z = X + Y$ is the sum of two independent random variables X and Y, each having a distribution $f_X(x)$ and $f_Y(y)$. Clearly, the probability density at $(X, Y) = (x, y)$ is the joint distribution

$$f_{XY}(x, y) = f_X(x) \cdot f_Y(y). \tag{3.4}$$

But what is the probability density for $Z = z$? We should "sum over" the probability densities $f_{XY}(x, y)$ for all the points (x, y) such that $y = z - x$, which is

$$f_Z(z) = \int dx f_{XY}(x, z - x) = \int dx f_X(x) f_Y(z - x). \tag{3.5}$$

This kind of integral is called the **convolution** of the two functions $f_X(x)$ and $f_Y(y)$, usually denoted as $f_Z = f_X * f_Y$.

It can be perceived that when there are more than two random variables, even though being independent, the distribution of their sum will be extremely complicated.

3.2 Making use of Fourier transform (characteristic function)

In some textbooks (e.g. [6]), it is said that Laplace found himself a way out by looking into the **moment generating function** (MGF) of every $f_i(x)$, defined as

$$M_i(t) = \mathbb{E}(e^{t\epsilon_i}) = \int dx\ e^{tx} f_i(x). \tag{3.6}$$

It bears its name because its k^{th} derivative evaluated at $t = 0$ is ϵ_i's k^{th} moment about 0: $\mathbb{E}(\epsilon_i^k)$. (We leave the proof as an exercise to the reader, and the solution can be found in any statistics textbook, or even in Wikipedia.) This story, though sounds very appealing because eq. (3.6) looks just like the two-sided **Laplace transform**[14] of $f_i(x)$ (with sign reversal in the exponential), is not only unreal in the history but also incorrect in mathematics[15].

The actual approach by Laplace [5] was to make use of the **characteristic function** of $f_i(x)$

$$\phi_i(t) = \mathbb{E}(e^{it\epsilon_i}) = \int dx\ e^{itx} f_i(x), \tag{3.8}$$

[14]For a function $f(x)$, its Laplace transform is defined as

$$\mathcal{L}[f](s) = \int_0^\infty dx e^{-sx} f(x), \tag{3.7}$$

and the two-sided version is simply extending the lower limit of the integral to $-\infty$.

[15]The problem is that the integral in eq. (3.6) is not always well-defined, even for some widely used distributions satisfying the central limit theorem –the final result of this chain of derivations. Some textbooks would argue that we only need $M_i(t)$ to be well defined in the a small neighborhood of $t = 0$. Well, indeed we can derive the desired result with that assumption, but still there are some distributions violating this assumption satisfy the central limit theorem, e.g. the log-normal distribution (for X being log-normally distributed, $\log X$ is normally distributed), whose MGF is infinite for any $t > 0$ and is only well-defined at $t \leq 0$.

which is the Fourier transform (with sign reversal) of $f_i(x)$ (see Appendix A.2 if you need more information). Since $|e^{itx}| < 1$ and $f_i(x)$ is non-negative,

$$|\phi(t)| \leq \int dx f_i(x) = 1, \tag{3.9}$$

so that $\phi(t)$ is guaranteed to be well-defined[16] at all t. The characteristic function eq. (3.8)'s k^{th} derivative is related to $\mathbb{E}(\epsilon_i^k)$ in the following way:

$$\phi_i^{(k)}(0) = i^k \mathbb{E}(\epsilon_i^k). \tag{3.10}$$

The proof, which can be found everywhere, is also left to the readers as an exercise.

Let's get back to the main stream: Laplace needed to find the distribution of a sum $\delta = \epsilon_1 + \dots \epsilon_N$ and he proposed making use of characteristic function. The characteristic function $\phi(t)$ of δ is

$$\begin{aligned}
\phi(t) &= \mathbb{E}(e^{it\delta}) = \mathbb{E}(e^{it(\epsilon_i + \dots + \epsilon_N)}) \\
&= \mathbb{E}(e^{it\epsilon_1} \cdot e^{it\epsilon_2} \dots e^{it\epsilon_N}) \\
&= \mathbb{E}(e^{it\epsilon_1}) \cdot \mathbb{E}(e^{it\epsilon_2}) \dots \mathbb{E}(e^{it\epsilon_N}) \\
&= \prod_{i=1}^{N} \phi_i(t),
\end{aligned} \tag{3.11}$$

where in the second to last equality we used the fact that all ϵ_i's are independent. Now, we see that though the distribution $f(x)$ of δ is hard to extract from the $f_i(x)$'s, the characteristic function $\phi(t)$ of δ is a simple product of all the $\phi_i(t)$'s. Today, taking the Fourier transform is the universally standard approach for dealing with convolutions. It is worth noting that Laplace's earlier use of the characteristic function captured exactly the same underlying intuition. He successfully applied this principle years before Fourier formally introduced his groundbreaking theory of series and transform in 1822 [7].

Whenever seeing a product, taking logarithm is almost always a good idea:

$$\log \phi(t) = \sum_{i=1}^{N} \log \phi_i(t). \tag{3.12}$$

To further simplify it, let's expand every $\phi_i(t)$ near $t = 0$:

$$\phi_i(t) = 1 - \frac{\sigma_i^2}{2!} t^2 + o(t^3), \tag{3.13}$$

[16] Here we only mention one of the three conditions for the existence of Fourier transform. Readers who need more information may search for "Dirichlet conditions" online.

where we have used the fact eq. (3.10) that $\phi_i'(0) = i\mathbb{E}(\epsilon_i) = 0$ and $\phi_i''(0) = i^2\mathbb{E}(\epsilon_i^2) = -\mathbb{V}(\epsilon_i) = -\sigma_i^2$. Therefore,

$$\begin{aligned}\log\phi(t) &= \sum_{i=1}^{N}\log\left(1-\frac{\sigma_i^2}{2}t^2+o(t^3)\right)\\ &\approx \sum_{i=1}^{N}\left(-\frac{\sigma_i^2}{2}t^2+o(t^3)\right)\\ &= -\frac{\sigma^2}{2}t^2+o(t^3),\end{aligned} \tag{3.14}$$

where we approximated $\log(1+x) \approx x$ for $x \ll 1$ and defined $\sigma^2 = \sum_{i=1}^{N}\sigma_i^2$. Next, omitting the higher order terms[17], we get

$$\log\phi(t) \approx -\frac{\sigma^2}{2}t^2, \tag{3.15}$$

which indicates

$$\phi(t) = e^{-\frac{\sigma^2}{2}t^2}. \tag{3.16}$$

Doing an inverse Fourier transform on $\phi(t)$ will give us the distribution $f(x)$ of δ:

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dt\ e^{-itx}\phi(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dt\ e^{-(itx+\frac{\sigma^2}{2}t^2)}. \tag{3.17}$$

To do the integral, we complete square on the exponential:

$$\frac{\sigma^2}{2}t^2+itx = \left(\frac{\sigma}{\sqrt{2}}t\right)^2+itx+\left(\frac{ix}{\sqrt{2}\sigma}\right)^2-\left(\frac{ix}{\sqrt{2}\sigma}\right)^2 = \left(\frac{\sigma}{\sqrt{2}}t+\frac{ix}{\sqrt{2}\sigma}\right)^2+\frac{x^2}{2\sigma^2}, \tag{3.18}$$

then

$$f(x) = \frac{1}{2\pi}e^{-\frac{x^2}{2\sigma^2}}\int_{-\infty}^{\infty} dt\ e^{-\left(\frac{\sigma}{\sqrt{2}}t+\frac{ix}{\sqrt{2}\sigma}\right)^2} \tag{3.19}$$

becomes a standard Gaussian integral, see eq. (A.5) in Appendix A.1. Our final result is

$$f(x) = \frac{1}{2\pi}e^{-\frac{x^2}{2\sigma^2}}\sqrt{\frac{2\pi}{\sigma^2}} = \frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{x^2}{2\sigma^2}}, \tag{3.20}$$

exactly the same as eq. (2.13) –a normal distribution!

3.3 The central limit theorem

This result by Laplace, called the **central limit theorem** (CLT), soon became the foundation of modern statistical inference and observational sciences. As Stigler famously argued in his authoritative [3], Laplace's 1810 (1812) work was a conceptual watershed. Before Laplace, Gauss had mathematically reverse-engineered the normal distribution in 1809 by

[17] If you doubt the validity of doing this, or even the approximations in the previous steps, you are not alone! We will come back to discuss them right after.

assuming the arithmetic mean to be the optimal estimator –a brilliant mathematical convenience, but one lacking a generative physical mechanism. Laplace, according to Stigler, provided exactly that mechanism, liberating the normal distribution of random errors from an epistemological consequence.

Stigler emphasizes that this insight transformed probability theory from a study of gambling games into an indispensable framework for interpreting complex astronomical and physical data. The normal distribution was no longer just an arbitrary hypothesis or a convenient mathematical trick; it was revealed as an inevitable result arising from the composition of independent error sources. This shift forever changed how scientists viewed uncertainty: complexity at the level of independent sources reliably dissolves into elegant predictability at the level of the total error observed.

To encapsulate this profound discovery into the rigorous language of modern probability theory, a formal statement of the CLT is needed here. However, we have to deal with a few technical issues before pursuing mathematical rigor.

3.4 Discussion on the series expansion and approximations

Astute readers will likely notice a mathematical gap in the derivation above: the series expansions and approximations spanning eq. (3.13) to eq. (3.15) lack rigorous justification. The procedure almost seems as if we deliberately truncated the series to reverse-engineer our target characteristic function, $\phi(t) = \exp(-\frac{\sigma^2}{2}t^2)$. The core issue is that we neglected terms of $o(t^3)$ and higher without any argument —a critical oversight considering the ensuing inverse Fourier transform requires integrating t from $-\infty$ to $+\infty$, a domain over which our local approximation completely breaks down.

The crucial insight that validates this treatment is Laplace's realization that the total number of error sources, N, must be immense. Consider the error δ arising from just three sources $\epsilon_{1,2,3}$; its distribution $f(x)$ would be highly sensitive to the specific forms of probability densities $f_{1,2,3}(x)$ of those individual components. Only in the limit of large N can the microscopic details of every distribution $f_i(x)$ possibly wash out, yielding a universal macroscopic outcome like the normal distribution.

However, blindly taking $N \to \infty$ would immediately lead us to trouble, for instance, if every variance σ_i^2 is finite, the total variance $\sigma^2 = \sum_{i=1}^{N} \sigma_i^2$ can easily diverge, ultimately resulting in an "infinitely flat" bell-shaped curve with no physical meaning. To get around this dilemma, we have to require the total variance σ^2 to be constant as $N \to \infty$. This makes perfect sense because σ^2 models the finite, measurable variance of the observations in real experiments.

Even though the summation $\sum_{i=1}^{N} \sigma_i^2$ does not diverge, there is another potential trouble associated with $N \to \infty$. Imagine one of the variances, say σ_1^2, is dominantly large[18]. It basically means that the random error δ obtained in an observation is mainly contributed by this particular source ϵ_1, therefore, the distribution $f(x)$ of δ will likely to resemble the distribution $f_1(x)$ of ϵ_1 –once again giving rise to a non-universal result. To avoid this problem, we need to enforce that none of the σ_i^2's dominant, that is, as $N \to \infty$, $\sigma_{\max}^2 = \max(\sigma_1^2, \ldots, \sigma_N^2) \to 0$.

The above two requirements basically render an idea that every σ_i shrinks down to zero as $N \to \infty$, but the reader may argue that the distribution $f_i(x)$ of an ϵ_i could have such a weird shape that its *absolute*[19] third moment $\mathbb{E}(|\epsilon_i^3|)$, which gives the order of magnitude of the coefficient of the t^3 term in eq. (3.13), never vanishes as $N \to \infty$, resulting in a non-negligible $\sim t^3$ term, ruining the nice result eq. (3.15). Indeed, it indicates that for the CLT to hold, additional requirements are needed to further restrict the distributions $f_i(x)$'s. We will state some precise mathematical conditions for CLT to hold in the appendix of this note, while keep a rough feeling here that every $f_i(x)$ should not possess a "thick tail". Actually, for most distributions we commonly see, their absolute third moments usually scale with the cube of the standard deviation[20], i.e., $\mathbb{E}(|\epsilon_i^3|) \sim \sigma_i^3$. In this way, preserving the $o(t^3)$ term we get

$$\log \phi(t) \approx -\frac{\sigma^2}{2} t^2 + \sum_{i=1}^{N} o(\sigma_i^3) t^3, \tag{3.21}$$

where

$$\sum_{i=1}^{N} \sigma_i^3 \leq \sigma_{\max} \sum_{i=1}^{N} \sigma_i^2 = \sigma_{\max} \sigma^2. \tag{3.22}$$

Clearly, it will vanish as $N \to \infty$ since $\sigma_{\max} \to 0$, so as the $o(t^3)$ term in $\log \phi(t)$, justifying the result in eq. (3.15).

As a conclusion, we provide a not-so-rigorous but practically useful mathematical statement of the central limit theorem, summarizing the main idea of the above discussion and reflecting the spirit of Laplace's original work.

Theorem 3.1 (Laplace's central limit theorem)**.** *Let $\epsilon_1, \epsilon_2, \ldots, \epsilon_N$ be a sequence of independent random variables, each with zero expectation and variance σ_i^2. As $N \to \infty$, if the total variance*

$$\sigma^2 = \sum_{i=1}^{N} \sigma_i^2 \tag{3.23}$$

converges and $\sigma_{max}^2 = \max(\sigma_1^2, \ldots, \sigma_N^2) \to 0$, as well as the distribution $f_i(x)$ of every ϵ_i does not have a thick tail, the sum $\sum_{i=1}^{N} \epsilon_i$ for a large enough N would be approximately normally distributed $\sim \mathcal{N}(0, \sigma^2)$.

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[18] As a not-so-insightful example, let's take $\sigma_i^2 = a_i \times 10^{-i}$, where a_i is a positive integer no larger than 9, then $\sigma^2 = \sum_{i=1}^{N} \sigma_i^2 = 0.a_1 a_2 a_3 \ldots a_N$ in decimals.

[19] We take the absolute value so that the positive and negative do not cancel, providing a looser bound of the estimation.

[20] For example, denoting the standard deviation as σ, the absolute third moment of an even distribution is $\frac{3\sqrt{3}}{4}\sigma^3$, that of a normal distribution is $\sqrt{\frac{8}{\pi}}\sigma^3$, that of a Laplace distribution is $\sqrt{\frac{9}{2}}\sigma^3$.

The conditions for the CLT to rigorously hold, as mentioned in Appendix A.3, are in fact abnormally complicated and thus understanding them is beyond the aim of this note. The take home message for readers is: as long as the distribution $f_i(x)$'s of all the summands ϵ_i's satisfy some conditions, the seemingly flawed approximations from eq. (3.13) to eq. (3.15) can rigorously go through, resulting in the normal distribution of the summation.

4 Bibliography

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A Appendix

A.1 Gaussian integral

The Gaussian integral, also known as the Euler–Poisson integral, is the integral of the Gaussian function $f(x) = e^{-x^2}$ over the entire real line:

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx. \tag{A.1}$$

The trick is to take the square and make it into a double integral

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx\right)^2 = \int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{\infty} e^{-y^2} dy = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dxdy, \tag{A.2}$$

where the double integral over the entire $\mathbb{R}^2$ plane can be re-expressed using polar coordinates: $r^2 = x^2 + y^2$ and $dxdy = rdrd\theta$:

$$I^2 = \int_0^{\infty} e^{-r^2} rdr \int_0^{2\pi} d\theta = \pi \int_0^{\infty} e^{-r^2} 2rdr = \pi \int_0^{\infty} e^{-r^2} d(r^2) = \pi, \tag{A.3}$$

therefore,

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}. \tag{A.4}$$

Integrals of similar kinds often appear in math and physics, which can be evaluated using the above method. The most generic case with two real parameters a, b is given below:

$$\int_{-\infty}^{\infty} dx\ e^{-a(x+b)^2} = \sqrt{\frac{\pi}{a}}, \tag{A.5}$$

which one can easily prove by change of variables. Alternatively,

$$\int_{-\infty}^{\infty} dx\ e^{-(ax^2-bx+c)} = \sqrt{\frac{\pi}{a}}\, e^{\frac{b^2}{4a}-c}. \tag{A.6}$$

It is not more generic because one can recover the form eq. (A.5) by completing square in the exponential.

A.2 Fourier transform and its inverse

To fully master the Fourier transform[21], a college student of math, statistics, physics, or other relevant majors would need to take a semester-long course titled complex analysis. Here, we try to provide a crash course on the subject, helping readers to quickly get a sense of it.

The standard definition of the forward Fourier transform of a function $f(x)$ is

$$F(t) = \int_{-\infty}^{\infty} f(x) e^{-itx} dx, \tag{A.7}$$

[21] Or more generically speaking: Fourier analysis.

where t is a real parameter[22]. The word "transform" indicates that the function $F(t)$, if well-defined on the entire real line ($\forall t \in \mathbb{R}$), contains exactly the same information as the original $f(x)$. That is, there is a one-to-one correspondence between a function $f(x)$ and its Fourier transform $F(t)$:

$$f(x) \xrightarrow{\mathcal{F}} F(t), \quad F(t) \xrightarrow{\mathcal{F}^{-1}} f(x), \tag{A.8}$$

where $\mathcal{F}$ denotes the Fourier transform that maps $f(x)$ to $F(t)$ and $\mathcal{F}^{-1}$ denotes its inverse[23].

The explicit form of the inverse Fourier transform $\mathcal{F}^{-1}$ reconstructing $f(x)$ from $F(t)$ is

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(t) e^{itx} dt. \tag{A.9}$$

To prove this, we start with the right-hand side (RHS) of the inverse equation and substitute our definition of $F(t)$ into it. To avoid confusing our variables of integration, we will use x' as the dummy variable for the inner integral:

$$\text{RHS} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(x') e^{-itx'} dx' \right] e^{itx} dt. \tag{A.10}$$

Assuming the function $f(x)$ is sufficiently well-behaved (e.g., it decays quickly at infinity), we can swap the order of integration. We pull $f(x')$ and the dx' integration to the outside, and group the exponential terms involving k on the inside:

$$\text{RHS} = \int_{-\infty}^{\infty} f(x') \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{it(x-x')} dt \right] dx'. \tag{A.11}$$

The term inside the brackets is the standard integral representation of the Dirac delta function[24], $\delta(x - x')$:

$$\delta(x - x') = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{it(x-x')} dt \tag{A.12}$$

Substituting the delta function back into our equation yields:

$$\text{RHS} = \int_{-\infty}^{\infty} f(x') \delta(x - x') dx'. \tag{A.13}$$

The defining characteristic of the Dirac delta function is its "sifting" property under an integral: it pulls out the value of the function exactly where the argument of the delta function is zero (which is at $x' = x$)

$$\text{RHS} = \int_{-\infty}^{\infty} f(x') \delta(x - x') dx' = f(x). \tag{A.14}$$

[22]Readers may realize that the convention here is to use e^{-itx}, whereas Laplace used e^{itx} in his definition of characteristic function. Actually, all the following derivations will go through if we replace all the i with $-i$, and readers should convince themselves about it.

[23]Other notations like $F(t) = \mathcal{F}[f(x)]$ are also widely used in literature.

[24]Conceptually, integrating an oscillating complex exponential over all space results in total destructive interference everywhere except where $x = x'$. At $x = x'$, the exponent becomes 0, $e^0 = 1$, and the integral sums to an infinite peak. More information can be found in Wikipedia.

This completes the proof of eq. (A.9). By substituting the forward transform into the inverse transform formula, we successfully recover the original function $f(x)$.

The Fourier transform $F(t)$ can be understood as a different perspective of seeing the original function $f(x)$. If you find it hard to deal with $f(x)$, moving to $F(t)$ may provide you with some insights[25].

A.3 The rigorous statements of central limit theorem

In this section, we provide a few famous mathematically rigorous statements of the central limit theorem, without any proof or understanding. Readers who are interested in the details may refer to advanced textbooks in statistics.

The first two statements give the conditions under which the CLT can rigorously hold in the $N \to \infty$ limit. They are usually seen as two different versions of the CLT.

Theorem A.1 (Lyapunov central limit theorem)**.** *Let $X_1, X_2, \ldots, X_N$ be a sequence of independent random variables, each with finite expectation μ_i and variance σ_i^2. Define*

$$s_N^2 = \sum_{i=1}^{N} \sigma_i^2, \tag{A.15}$$

If for some $\delta > 0$,

$$\lim_{N\to\infty} \frac{1}{s_N^{2+\delta}} \sum_{i=1}^{N} \mathbb{E}\left[|X_i - \mu_i|^{2+\delta}\right] = 0, \tag{A.16}$$

*(this is called **Lyapunov's condition**) then, as $N \to \infty$, the normalized sum of these random variables converges in distribution to the standard normal distribution $\mathcal{N}(0,1)$:*

$$\frac{1}{s_N} \sum_{i=1}^{N} (X_i - \mu_i) \xrightarrow{d} \mathcal{N}(0,1). \tag{A.17}$$

The Lyapunov condition above can be replaced with a weaker one, leading to the Lindeberg-Feller CLT:

Theorem A.2 (Lindeberg-Feller central limit theorem)**.** *Let $X_1, X_2, \ldots, X_N$ be a sequence of independent random variables, each with finite expectation μ_i and variance σ_i^2. Define*

$$s_N^2 = \sum_{i=1}^{N} \sigma_i^2, \tag{A.18}$$

If for every $\varepsilon > 0$,

$$\lim_{N\to\infty} \frac{1}{s_N^2} \sum_{i=1}^{N} \mathbb{E}\left[(X_i - \mu_i)^2 \cdot \mathbf{1}_{\{|X_i-\mu_i|>\varepsilon s_N\}}\right] = 0, \tag{A.19}$$

[25] As a digression, in modern functional analysis we think of a function f as a vector in a infinite-dimensional linear space. From this perspective, $f(x)$ is a representation of the vector f in a basis, like the components (x, y, z) of a three-dimensional vector in a coordinate system, while $F(t)$ is the representation of the same vector f in a different basis.

(this is called ***Lindeberg's condition****, in which $\mathbf{1}_{\{|X_i-\mu_i|>\varepsilon s_N\}}$ is an indicator function which is 1 when $|X_i-\mu_i|>\varepsilon s_N$ and 0 otherwise) then, as $N\to\infty$, the normalized sum of these random variables converges in distribution to the standard normal distribution $\mathcal{N}(0,1)$:*

$$\frac{1}{s_N}\sum_{i=1}^{N}(X_i-\mu_i)\xrightarrow{d}\mathcal{N}(0,1). \tag{A.20}$$

The condition $N\to\infty$ for CLT to hold is sometimes impractical in real-world applications. Below we state two versions of the Berry-Esseen theorem, which is more practical than CLT under stronger assumptions.

Theorem A.3 (Berry-Esseen theorem (i.i.d. version))**.** *Let $X_1, X_2, \ldots, X_N$ be a sequence of independent and identically distributed (i.i.d.) random variables with expectation μ, variance $\sigma^2>0$, and finite third absolute central moment $\rho=\mathbb{E}\left[|X_1-\mu|^3\right]$.*

Let $F_N(x)$ be the cumulative distribution function of the normalized sum

$$Z_N=\frac{1}{\sigma\sqrt{N}}\sum_{i=1}^{N}(X_i-\mu), \tag{A.21}$$

and let $\Phi(x)$ be the cumulative distribution function of the standard normal distribution.

Then, there exists a universal constant $C>0$ such that for all $N\geq 1$ and for all $x\in\mathbb{R}$,

$$\sup_{x\in\mathbb{R}}|F_N(x)-\Phi(x)|\leq\frac{C\cdot\rho}{\sigma^3\sqrt{N}}. \tag{A.22}$$

Theorem A.4 (Berry-Esseen theorem)**.** *Let $X_1, X_2, \ldots, X_N$ be a sequence of independent random variables with expectation μ_i, variance $\sigma_i^2>0$, and finite third absolute central moment $\rho_i=\mathbb{E}\left[|X_i-\mu_i|^3\right]$.*

Define the total variance $s_N^2=\sum_{i=1}^{N}\sigma_i^2$, and let $F_N(x)$ be the cumulative distribution function of the normalized sum

$$Z_N=\frac{1}{s_N}\sum_{i=1}^{N}(X_i-\mu_i). \tag{A.23}$$

Let $\Phi(x)$ be the cumulative distribution function of the standard normal distribution.

Then, there exists a universal constant $C>0$ such that for all $N\geq 1$ and for all $x\in\mathbb{R}$,

$$\sup_{x\in\mathbb{R}}|F_N(x)-\Phi(x)|\leq C\cdot\frac{\sum_{i=1}^{N}\rho_i}{s_N^3}. \tag{A.24}$$